

Enrollment-based cohort strata in PPMI: governing protocols and inclusion/exclusion criteria

Strata are defined solely by enrollment-time information: the protocol version under which the participant was screened (PROTOCOL_ERA; from SCREENEDAM for PPMI Clinical protocol 002, otherwise PATNO allocation) crossed with the enrollment cohort. Criteria below are condensed summaries in the authors' words; the cited protocol documents (ppmi-info.org, Research Documents & SOPs and archive pages; ClinicalTrials.gov NCT01141023 and NCT04477785) remain the authoritative sources and should be consulted for the complete text, including cohort-specific medication washouts and female-of-childbearing-potential requirements common to all strata.

Enrollment stratum (PROTOCOL_ERA cohort)	Governing protocol document (citation for criteria)	Enrollment period; n enrolled (n consented)*	Key inclusion criteria (summarized)	Key exclusion criteria (summarized)	Biomarker eligibility gate	Identification in Participant_Status
PPMI1_Original PD (de novo sporadic)	PPMI protocol 001 (NCT01141023), original protocol & early amendments; Marek et al. 2018, Ann Clin Transl Neurol 5:1460-77	2010–2013 n = 423 (489)	Age ≥30 y; PD diagnosis ≤2 y before screening; untreated for PD and not expected to need PD medication within 6 months of baseline; ≥2 of resting tremor / bradykinesia / rigidity (incl. tremor or bradykinesia), or a single asymmetric resting tremor or asymmetric bradykinesia; H&Y I–II.	Prior/current dopaminergic therapy beyond minimal exposure; atypical or secondary parkinsonism; dementia; MRI evidence of other significant neurological disease; dopamine-blocking drugs within 6 months; LP contraindications.	Screening DaTscan SPECT read as showing DAT deficit (participants meeting clinical criteria without deficit were assigned to SWEDD).	PATNO 3000–4999 (excl. 5000s); SCREENEDAM missing; COHORT=1; ENLSRDC=1 (back-filled sporadic flag).
PPMI1_Original HC	PPMI protocol 001 (NCT01141023); Marek et al. 2018	2010–2013 n = 196 (241)	Age ≥30 y; no clinically significant neurological disorder; MoCA >26.	First-degree relative with PD; significant neurological disorder; dopamine-blocking or DAT-interfering medications; LP contraindications.	Screening DaTscan visually normal.	PATNO 3000–4999; SCREENEDAM missing; COHORT=2.
PPMI1_Original SWEDD	PPMI protocol 001 (NCT01141023); Marek et al. 2018	2010–2013 n = 64 (81)	Met all clinical PD-cohort criteria above.	As PD cohort.	Screening DaTscan WITHOUT dopaminergic deficit (defining feature). Legacy cohort; follow-up largely truncated except scan-positive re-invitations (protocol 001 Amendment 8, 12-May-2014).	PATNO 3000–4999; SCREENEDAM missing; COHORT=3.
PPMI1_GeneticProdromal PD (Genetic Cohort / Registry: LRRK2, GBA, SNCA)	PPMI protocol 001 amendments (Genetic Cohort & Genetic Registry arms; see NCT01141023 amendment history AM7 14-Oct-2013 through AM12 01-Jun-2017; archived PPMI-1 protocol on ppmi-info.org / LONI)	2013–2019 n = 280 (301)	Confirmed pathogenic LRRK2 / GBA / SNCA variant (or, for Registry, a first-degree relative of a carrier) with a PD diagnosis. NOT restricted to de novo disease: longer disease duration and ongoing PD treatment were permitted, unlike the original PD cohort. Recruitment enriched via founder populations and family-based outreach (Foroud et al. 2015).	General medical/LP exclusions analogous to core protocol; no DAT-deficit requirement for genetic arms.	None (genetic confirmation was the gate; DAT imaging acquired but not an eligibility criterion). GBA testing initially N370S only, broadened under protocol 001 Amendment 11 (01-Apr-2016).	PATNO 10000–99999; SCREENEDAM missing; COHORT=1 with ENRLRRK2 / ENRLGBA / ENRLSNCA = 1.
PPMI1_GeneticProdromal Prodromal (hyposmia / RBD pilot; genetic non-manifest)	PPMI protocol 001 amendments (Prodromal arm added 2013; NCT01141023 §4.2.7)	2013–2019 n = 494 (661)	No PD diagnosis, plus ≥1 of: (i) hyposmia — UPSIT at or below the 10th percentile for age and sex (olfactory core confirmation); (ii) polysomnography-confirmed RBD; (iii) documented LRRK2 (later GBA/SNCA) variant. Age ≥60 y for hyposmia/RBD arms. Enrollment targeted ~80% with DAT deficit and	Clinical PD, other parkinsonism, or dementia; dopamine-blocking medications; LP contraindications.	Imaging-core-read screening DaTscan required for hyposmia/RBD arms (DAT-deficit-enriched selection). Genetic non-manifest carriers	PATNO 10000–99999; SCREENEDAM missing; COHORT=4 with ENRLHPSM / ENRLRBD / ENRLRRK2 / ENRLGBA / ENRLSNCA = 1.

Enrollment stratum (PROTOCOL_ERA cohort)	Governing protocol document (citation for criteria)	Enrollment period; n enrolled (n consented)*	Key inclusion criteria (summarized)	Key exclusion criteria (summarized)	Biomarker eligibility gate	Identification in Participant_Status
			~20% comparable participants without deficit.		enrolled irrespective of DAT status.	
EarlyImaging_AV133 PD	PPMI Early Imaging protocol 004 (NCT04507139; archived Early Imaging protocol/AM2 08-Feb-2023 on ppmi-info.org)	2019–2020 n = 16 (19)	Untreated, recently diagnosed PD meeting core PPMI motor criteria; consented to dual DaTscan + [18F]AV-133 VMAT2 PET imaging. For most analyses poolable with de novo sporadic PD, but the imaging schedule differs.	As PD cohort, plus PET-specific contraindications and pregnancy exclusions.	DaTscan-based eligibility; additional AV-133 PET imaging.	PATNO 5000–5999; SCREENEDAM missing; AV133STDY=1; COHORT=1.
PPMI2_AM2 PD (sporadic + genetic subgroups)	PPMI Clinical protocol 002, Amendment 2, v1.2, 01-Feb-2021, §7.2–7.4 (also covers screens under original protocol 08-Apr-2020 / AM1 23-Jun-2020)	2020–2023 (few thereafter) n = 502 (723)	Sporadic (§7.2): age ≥30 y; PD diagnosis ≤2 y; untreated and not expected to need PD medication for 6 months post-baseline; core motor criteria; H&Y I–II. LRRK2/GBA (§7.3): same duration/staging but treatment status not restricted; Screening-Core-confirmed causative variant. SNCA/rare incl. PRKN/PINK1 (§7.4): no duration limit; H&Y I–III.	Sporadic: any levodopa/dopamine-agonist/MAO-B/amantadine use currently or within 60 d of baseline, or >90 d lifetime levodopa/agonist; atypical or secondary parkinsonism; dementia; dopamine-blocking drugs within 6 months; MRI evidence of other significant neurological disease; LP contraindications.	Screening DaTscan SPECT eligibility (pre-specified imaging cut-offs) required for all PD subgroups.	SCREENEDAM=2; COHORT=1; ENRLSRDC / ENRLRRK2 / ENRLGBA / ENRLSNCA / ENRLPRKN / ENRLPINK1 flags.
PPMI2_AM2 HC	Protocol 002 AM2, §7.1. NOTE: the original 08-Apr-2020 protocol set HC age ≥57 y; AM2 lowered it to ≥30 y — both screen waves carry SCREENEDAM=2.	2020–2023 n = 75 (94)	Age ≥30 y (≥57 y if screened under original 2020 protocol); able to consent; DAT-interfering medications must be washable.	First-degree relative with PD; significant neurological disorder (clinical or on prior MRI); dopamine-blocking drugs within 6 months; LP contraindications.	Screening DaTscan eligibility (normal-range imaging).	SCREENEDAM=2; COHORT=2.
PPMI2_AM2 Prodromal	Protocol 002 AM2, §7.5	2020–2023 n = 645 (1,194)	Centrally determined predictive eligibility (UPSIT-based, via PPMI Online → Remote funnel, or site-identified risk: first-degree relative, RBD, genetic variant); age ≥60 y (≥30 y for SNCA/rare variants); two-stage consent (screening, then full protocol).	Clinical PD, other parkinsonism, or dementia; PD medications currently or within 60 d of baseline; dopamine-blocking drugs within 6 months; LP contraindications.	Screening DaTscan gate: all with DAT deficit eligible (~75% of continuers expected); a selected minority without deficit (~25%; reduced-but-normal binding and/or high-risk RBD/genetic) also continued. DAT status masked to participant and site.	SCREENEDAM=2; COHORT=4; ENRLHPSM / ENRLRBD / genetic flags; DATELIG records imaging eligibility.
PPMI2_AM3 PD / HC / Prodromal	PPMI Clinical protocol 002, Amendment 3.2, v2.2, 30-Jan-2023, §7.1–7.5 (doi:10.17504/protocols.io.n92ldmw6ol5b/v2)	2023–2025 PD n = 231 (346); HC n = 64 (97); Prodromal n = 1,079 (2,414)	Substantively the same structure as AM2 (HC ≥30 y DaTscan-gated; sporadic PD de novo ≤2 y, H&Y I–II, DaTscan-gated; genetic PD subgroups; prodromal ≥60 y except SNCA/rare ≥30 y, UPSIT-based predictive criteria). Main changes vs AM2: low-dose RLS treatment allowed by medical-monitor exception; screening may add blood/skin research samples;	As AM2, with the RLS exception noted; otherwise unchanged in substance.	DaTscan gate retained for all cohorts; prodromal continuation still DAT-deficit-enriched with ~25% expected without deficit.	SCREENEDAM=3; COHORT + ENRL flags as above.

Enrollment stratum (PROTOCOL_ERA cohort)	Governing protocol document (citation for criteria)	Enrollment period; n enrolled (n consented)*	Key inclusion criteria (summarized)	Key exclusion criteria (summarized)	Biomarker eligibility gate	Identification in Participant_Status
			investigator-discretion catch-all exclusion added; predictive prodromal criteria continue to be iteratively optimized centrally.			
PPMI2_AM4 PD (sporadic, genetic, and NEW Normosmic arms)	PPMI Clinical protocol 002, Amendment 4, v3.0, 23-Jul-2024, §7.2–7.7 (participants consented under AM3.2 finish screening under Appendix 1 criteria)	2024/2025–ongoing n = 242 (293)	All PD: age ≥30 y; core motor criteria; central UPSIT eligibility required (i.e., hyposmia-enriched pathway). Sporadic (§7.3) and LRRK2/GBA (§7.4): diagnosis ≤2 y, H&Y I–II, SAA-status gate. SNCA/rare (§7.5): H&Y I–III, SAA gate. NEW PD Normosmic (§7.6) and PD Normosmic LRRK2 (§7.7): diagnosis ≤7 y, H&Y I–III, no SAA gate — a deliberately distinct, longer-duration, normosmic PD population.	Sporadic: PD-treatment exclusions as before (with RLS/DBS exceptions); atypical/secondary parkinsonism; dementia; dopamine-blocking drugs; LP contraindications.	Eligibility anchored on asyn SAA status (CSF, blood, skin, or other validated measure) for sporadic/genetic arms; DaTscan retained as an assessment but no longer the eligibility anchor.	SCREENEDAM=4; ENRLNORM=1 identifies the Normosmic arms; other ENRL flags as above.
PPMI2_AM4 HC	Protocol 002 AM4, §7.1	2024/2025–ongoing n = 1 (4; recruitment de-emphasized)	Age ≥30 y; central UPSIT eligibility; SAA-status gate to proceed to baseline.	First-degree relative with PD; significant neurological disorder; dopamine-blocking drugs; LP contraindications.	UPSIT (centrally determined) + asyn SAA status; SPECT no longer the screening gate.	SCREENEDAM=4; COHORT=2.
PPMI2_AM4 Prodromal	Protocol 002 AM4, §7.8	2024/2025–ongoing n = 739 (1,683)	Central UPSIT eligibility (hyposmia-anchored funnel); age ≥40 y (≥30 y for SNCA/rare variants) — lowered from ≥60 y; asyn SAA-status eligibility confirmation; able to consent.	Clinical PD, other parkinsonism, or dementia at screening; PD medications currently or within 60 d of baseline (RLS exception); dopamine-blocking drugs within 6 months; LP contraindications.	asyn SAA status (CSF, blood, skin, or other validated measure) replaces the DAT-deficit gate. Consistent with this, RBD-flagged enrollment effectively ceases (n=1 enrolled) and hyposmia becomes the dominant arm. Note: AM4 screening SAA is an eligibility variable; AM4 participants do not (yet) appear in the released research SAA results file (projects 155/237).	SCREENEDAM=4; COHORT=4; ENRLHPSM dominant; DATELIG sparsely populated.

* n enrolled = ENROLL_STATUS in {Enrolled, Withdrew, Complete, Withdraw Deceased, Baseline, Baseline Withdraw}; n consented additionally includes screen failures, exclusions, declines, and pending screens, all of whom are retained in the analytic dataset (Participant_Status download of 11 Aug 2026). PPMI2_AM_unknown (n=386 consented, 1 enrolled) comprises PATNO ≥100000 records without a recorded SCREENEDAM (predominantly pending or pre-screen dispositions) and is omitted from the table.

Abbreviations: AM, amendment; asyn SAA, alpha-synuclein seed amplification assay; DAT, dopamine transporter; H&Y, Hoehn & Yahr; HC, healthy control; LP, lumbar puncture; RBD, REM sleep behavior disorder; RLS, restless legs syndrome; UPSIT, University of Pennsylvania Smell Identification Test.